PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Relationship between osteonecrosis and antiphospholipid antibodies in patients with systemic lupus erythematosus: A systematic review protocol
AUTHORS	Wei, Qijiao; Zhou, Meng; Liu, Jianwen; Zhang, Shengli; Gao, Fei; LIn, He; Chen, Zhihan

VERSION 1 – REVIEW

REVIEWER	Wu, Zhenbiao Fourth Military Medical University
REVIEW RETURNED	01-Feb-2021

GENERAL COMMENTS	The abstract has not the part of results and conclusions. The paper also is in short of results.

REVIEWER	Ugarte-Gil, MF Universidad Cientifica del Sur
REVIEW RETURNED	06-Mar-2021

GENERAL COMMENTS	The manuscript is well-written, however, some clarifications are needed. How are you planning to analyze the impact of antiphospholipid antibodies? Per antibody, any antibody, any titer? Should the manuscripts have a minimal follow-up since the measurement of the a...

REVIEWER	Gran, Sonia University of Nottingham, Centre of Evidence Based Dermatology
REVIEW RETURNED	24-Mar-2021

GENERAL COMMENTS	1) Please can a native English speaker proof read this manuscript as different tenses are used and there are quite a few sentences which are not clear. For example: what are the presence of aPLs associated with osteonecrosis in SLE? SLE patients without limited to age will be included 2) Can more information be provided on the diagnostic criteria (page 6, line 14) 3) Why were participants with certain conditions excluded? (page 6, lines 14-20) 4) Why is sample size mentioned in the outcome assessment section? And what is the outcome for the odds ratio. Is it ON? (page 6, line 24) 5) Why are cross-sectional studies not included?

	6) Please can you make the research question clearer (page 5, line 50): what are the presence of aPLs associated with osteonecrosis in SLE 7) This sentence needs to be clearer as it is key to the inclusion criteria: the coexistence of ON, SLE, and aPLs (page 6, line 57) 8) Has the literature search been conducted? If so, please provide dates. 9) Has the analysis been done? This sentence makes it sound like is has: The raw numerical data were extracted from the studies. Meta-analysis was performed using Stata 11.0 software. (Page 7, lines 29-31). 10) What type of meta-analysis model will be conducted? Will it be DerSimonian and Laird? 11) Please can you clarify this sentence (Page 7, lines 31-35): Before combining the statistics, tests for heterogeneity will be conducted. If p>0.1, the fixed-effect model will be used to combine the data. If p<0.1, the random-effect model will be used. What heterogeneity test will be conducted? Normally a random-effects model is conducted. 12) What variables will be used for subgroup analyses? 13) Why is there a discussion and conclusion if the work has not been conducted?

VERSION 1 – AUTHOR RESPONSE

Reviews' Comments to Author:

Reviewer: 1 [Editor: it appears this reviewer did not realize that it was a protocol paper]
Dr. Zhenbiao Wu, Fourth Military Medical University
Comments to the Author:
The abstract has not the part of results and conclusions.
The paper also is in short of results .
Thanks for your comments. My manuscript is a systematic review protocol, so there are no results and conclusions in our abstract. And there is no results in our manuscript. I hope I have got your comments, thanks.

Reviewer: 2
Dr. MF Ugarte-Gil, Universidad Cientifica del Sur
Comments to the Author:
The manuscript is well-written, however, some clarifications are needed.
1) How are you planning to analyze the impact of antiphospholipid antibodies? Per antibody, any antibody, any titer?
Thanks for your comments. We want to assess the relationship between antiphospholipid antibodies (aPLs) and the incidence of osteonecrosis (ON) in systemic lupus erythematosus (SLE) patients. aPLs are including lupus anticoagulant (LA), anticardiolipin (ACL) and antiβ2GP1 antibodies. We will analyze the relationship between three antibodies and ON respectively. Analysis about each isotype (IgG, IgM) of ACL, (IgG, IgM) anti-B2GPI antibodies should be performed. Through the literature, we find the titer of the antibody in each study is different. So each study has its own positive standard. What we will do is to sum up how many patients have positive aPLs in ON group and Non-ON group respectively. Finally, we will compare the positive rates between two groups if possible. I hope my answer can satisfy you, thanks.

2) Should the manuscripts have a minimal follow-up since the measurement of the antibody to the presence of the osteonecrosis or the end of the follow-up?"
Thanks for your comments. We should have a minimal follow-up since the measurement of the antibody to the presence of the osteonecrosis, since the measurement of the antibody to the end of the follow-up, since the presence of the osteonecrosis to the end of the follow-up. These contents will be added in Data extraction section. I hope I have got your comments, thanks.

Reviewer: 3
Dr. Sonia Gran, University of Nottingham
Comments to the Author:
1) Please can a native English speaker proof read this manuscript as different tenses are used and there are quite a few sentences which are not clear. For example:
what are the presence of aPLs associated with osteonecrosis in SLE?
SLE patients without limited to age will be included
Thanks for your comments. The correct statement of the first sentence is “What is the relationship between aPLs and osteonecrosis in SLE?”. The second one is “SLE patients without age limitation will be included”. I have modified them in the manuscript.
A native English speaker has polished our manuscript.
2) Can more information be provided on the diagnostic criteria (page 6, line 14)
Cases have clear diagnostic criteria. The diagnosis of SLE should have been made according to the 1982 revised American Rheumatism Association criteria, 1997 American College of Rheumatology classification criteria, Systemic Lupus International Collaborating Clinics classification criteria, or 2019 SLE European League Against Rheumatism/American College of Rheumatology classification criteria. ON should have been radiologically confirmed using plain radiography, bone scans, magnetic resonance imaging, or tomography.
3) Why were participants with certain conditions excluded? (page 6, lines 14-20)
Patients with other rheumatology diseases and traumatic osteonecrosis or infectious osteonecrosis will be excluded. Because we want to explore the relationship between aseptic ON and different aPLs just in SLE. The pathogenesis of traumatic osteonecrosis or infectious osteonecrosis and aseptic ON are different.

4) Why is sample size mentioned in the outcome assessment section? And what is the outcome for the odds ratio. Is it ON? (page 6, line 24)
①We will sum up the number of patients who have positive aPLs in ON group and Non-ON group respectively, then to compare the positive rates between two groups if possible. So it is important to provide clear regulations on the sample size in the studies.
②In literatures, logistic regression analyses will be performed to examine the risk factor for ON. aPLs may be the risk factor. So it will provide the odd ratio (OR) and the 95% confidence interval (95% CI) for positivity of any aPLs.
5) Why are cross-sectional studies not included?
Cross-sectional studies will be included. I have added the content in the manuscript (Type of studies section).

6) Please can you make the research question clearer (page 5, line 50): what are the presence of aPLs associated with osteonecrosis in SLE
The correct statement is “What is the relationship between aPLs and osteonecrosis in SLE?”.

7) This sentence needs to be clearer as it is key to the inclusion criteria: the coexistence of ON, SLE, and aPLs (page 6, line 57)
The correct statement is “Studies pertaining to ON, SLE, and aPLs will be included.”.

8) Has the literature search been conducted? If so, please provide dates.

We didn't conduct the literature search before submission of this manuscript on 25 November 2020. We did that on 26 December 2020. I have added the content in the manuscript (Search methods for identification of studies section).Like this: "The following six databases will be searched from their inception up until 26 December 2020. "

9) Has the analysis been done? This sentence makes it sound like is has:
The raw numerical data were extracted from the studies. Meta-analysis was performed using Stata 11.0 software. (Page 7, lines 29-31).
The correct statement is "Raw numerical data will be extracted from the studies. A meta-analysis will be performed, if possible, using Stata 11.0 software."

10) What type of meta-analysis model will be conducted? Will it be DerSimonian and Laird?
If $p < 0.1$, the DerSimonian and Laird random-effect model will be used..

11) Please can you clarify this sentence (Page 7, lines 31-35): Before combining the statistics, tests for heterogeneity will be conducted. If $p>0.1$, the fixed-effect model will be used to combine the data. If $p<0.1$, the random-effect model will be used. What heterogeneity test will be conducted? Normally a random-effects model is conducted.
Before combining the statistics, we will conduct heterogeneity tests using Stata 11.0 software. Heterogeneity test in Meta-analysis can be completed by the four methods(Q test, I2 statistic, H statistic and Galbraith plot) in Stata 11.0 software. In our study, we will use I2 statistic method. If $p>0.1$, a fixed-effect model will be used. If $p<0.1$, a random-effect model will be used. And I agree with your opinion: normally a random-effects model is conducted.

12) What variables will be used for subgroup analyses?
A subgroup analysis will be performed according to the study design type, the country of origin of the participants, their age, and sample size if necessary.

13) Why is there a discussion and conclusion if the work has not been conducted?
According to the Preferred Reporting Items for Systematic review and Meta-Analysis Protocols (PRISMA-P), discussion is not included. While, I just explain the reason and the background I do this protocol in our manuscript. I think it is necessary to tell readers the research question sufficiently.

VERSION 2 – AUTHOR RESPONSE

Comments from reviewer：

- Reviewer 2, comment 1

- please add some detail to the manuscript Methods section of how you intend to address the Ab impact.

Thanks for your comments. The following sentence is added in METHODS AND ANALYSIS＞ Research question: We only explore the relationship between the positive rate of aPLs and ON, not involving the titer of aPLs. See page 4 line 1-2.

- Please rename the Article Summary section to "Strengths and Limitations".

Thanks for your comments. I have changed "Article Summary" to "Strengths and Limitations". See page 2 the penultimate line.